Infixation on the CV Tier: Evidence from Alabama Agent Agreement

Supplementary Handout, May 1 2026

romihill.github.io/assets/files/bmw_poster.pdf
romihill.github.io

1 Relevant Rules

(1) **Vocabulary Insertion Rules**

1. $\text{AGR}_{[2,\ \text{SG},\ \text{agent}]} \longleftrightarrow$ *ch*
2. $\text{AGR}_{[2,\ \text{PL},\ \text{agent}]} \longleftrightarrow$ *hach*
3. $\text{AGR}_{[1,\ \text{PL},\ \text{agent}]} \longleftrightarrow$ *hili* / two-mora final foot
4. $\text{AGR}_{[1,\ \text{PL},\ \text{agent}]} \longleftrightarrow$ *l* / elsewhere
5. $\text{AGR}_{[1,\ \text{SG},\ \text{agent}]} \longleftrightarrow$ *a* / _ NEG
6. $\text{AGR}_{[1,\ \text{SG},\ \text{agent}]} \longleftrightarrow \varnothing$ / elsewhere
7. NEG $\longleftrightarrow$ *k*
8. AUX $\longleftrightarrow$ *o*

(2) **Pivot Rules**

- If right-most V is filled, then:
  - Insert the infix between the first $\text{V}_{i,[+\text{FILLED}]}$ $\text{C}_{i+1,[+\text{FILLED}]}$ sequence from the right. Empty CV slots are ignored.
    * Agent agreement marker is inserted to the left of $\text{C}_{i+1,[+\text{FILLED}]}$.
    * Negation marker is inserted to the right of $\text{V}_{i,[+\text{FILLED}]}$.
  - If there is no instance of $\text{V}_{i,[+\text{FILLED}]}$ $\text{C}_{i+1,[+\text{FILLED}]}$, place the agreement marker to the left of the left-most filled slot (i.e. prefixing iCV verbs).
- Else (i.e. right-most V is unfilled):
  - Insert the <C V> material at the end of the word (i.e. as a suffix).

(3) **Phonological Rules**

- Ungoverned V slots undergo *i*-insertion.
- *i*-insertion applies to monosyllabic roots.
- *i*-insertion applies to consonant-final stems.
- Complex segments (i.e. *ch*) must be in a licensed C.
- V $\rightarrow \varnothing$ / __ V

2 Positive Agreement

TAM morphology is ignored, mostly for space reasons.

2.1 CVVCV

hinaata $\rightarrow$ *hinaa<s>ta*

(4) a. Underlying form *hinaata*

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4
h	i	n	a			t	a

b. Insert material containing *ch* to the left of C_4

C_1	V_1	C_2	V_2	C_3	V_3	<C_5	V_5>	C_4	V_4
h	i	n	a			ch		t	a

c. *ch* is ungoverned and unlicensed, spirantization to *s*

C_1	V_1	C_2	V_2	C_3	V_3	<C_5	V_5>	C_4	V_4
h	i	n	a			s		t	a

d. *a* spreads to V_3

C_1	V_1	C_2	V_2	C_3	V_3	<C_5	V_5>	C_4	V_4
h	i	n	a			s		t	a

2.2 CVCCV

hokcho → *ho<chi>kcho*

(5) a. Underlying form *hokcho*

C_1	V_1	C_2	V_2	C_3	V_3
h	o	k		ch	o

b. Insert material containing *ch* before C_2

C_1	V_1	$<C_4$	$V_4>$	C_2	V_2	C_3	V_3
h	o	ch		k		ch	o

c. V_4 is ungoverned, insert *i*

C_1	V_1	$<C_4$	$V_4>$	C_2	V_2	C_3	V_3
h	o	ch	i	k		ch	o

2.3 CVCV

hompani → hompanchi

- In CVCV verb, the final V in the timing tier is not associated with its melodic material. It links only when the agent agreement marker is not present.

(6) a. Underlying form *hompani*

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4
h	o	m	a	p	a	n	i

b. Insertion of agreement marker *hompanch*

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4	C_5	V_5
h	o	m	a	p	a	n	i	ch	

c. Insertion of *i* in word final position

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4	C_5	V_5
h	o	m	a	p	a	n	i	ch	i

2.4 iCV Verbs

ipa $\rightarrow$ *ilpa*

(7) a. Underlying form *pa*

C_1	V_1	C_2	V_2
		p	a

b. Insert *ch* material to the left of C_2

C_1	V_1	<C_3	V_3>	C_2	V_2
		l		p	a

c. *i* is inserted into V_1 (no monosyllabic stems)

C_1	V_1	<C_3	V_3>	C_2	V_2
	i	l		p	a

2.5 *-ka* Verbs

(8) Underlying form *-ka*

C_1	V_1	C_2	V_2
	∅	k	a

loska $\rightarrow$ *los<is>ka*

(9) a. Underlying form *loska*

$$\begin{array}{cccccccc} C_1 & V_1 & C_2 & V_2 & C_3 & V_3 & C_4 & V_4 \\ | & | & | & & & | & | & | \\ \mathrm{l} & \mathrm{o} & \mathrm{s} & & & \varnothing & \mathrm{k} & \mathrm{a} \end{array}$$

b. Insert *ch* material to the left of C_4

$$\begin{array}{cccccccccc} C_1 & V_1 & C_2 & V_2 & C_3 & V_3 & <C_5 & V_5> & C_4 & V_4 \\ | & | & | & & & | & | & & | & | \\ \mathrm{l} & \mathrm{o} & \mathrm{s} & & & \varnothing & \mathrm{ch} & & \mathrm{k} & \mathrm{a} \end{array}$$

c. *Empty CV segment carrying ∅ deletes at some stage of the derivation*

$$\begin{array}{cccccccc} C_1 & V_1 & C_2 & V_2 & <C_5 & V_5> & C_4 & V_4 \\ | & | & | & & | & & | & | \\ \mathrm{l} & \mathrm{o} & \mathrm{s} & & \mathrm{ch} & & \mathrm{k} & \mathrm{a} \end{array}$$

d. V_2 is ungoverned, so we insert a *i* to it

$$\begin{array}{cccccccc} C_1 & V_1 & C_2 & V_2 & <C_5 & V_5> & C_4 & V_4 \\ | & | & | & | & | & & | & | \\ \mathrm{l} & \mathrm{o} & \mathrm{s} & \mathrm{i} & \mathrm{ch} & & \mathrm{k} & \mathrm{a} \end{array}$$

e. Spirantization, since C_3 is neither governed nor licensed

$$\begin{array}{cccccccc} C_1 & V_1 & C_2 & V_2 & <C_5 & V_5> & C_4 & V_4 \\ | & | & | & | & | & & | & | \\ \mathrm{l} & \mathrm{o} & \mathrm{s} & \mathrm{i} & \mathrm{s} & & \mathrm{k} & \mathrm{a} \end{array}$$

2.6 *-li* Verbs

bitli → *bitchi*

(10) a. Underlying form *bitli*

$$\begin{array}{cccc} C_1 & V_1 & C_2 & V_2 \\ | & | & | & \\ \mathrm{b} & \mathrm{i} & \mathrm{t} & \end{array}$$

b. Insert after V_2, since right-most V is unfilled

$$\begin{array}{cccccc} C_1 & V_1 & C_2 & V_2 & C_3 & V_3 \\ | & | & | & & | & \\ \mathrm{b} & \mathrm{i} & \mathrm{t} & & \mathrm{ch} & \end{array}$$

c. *i*-insertion at end of word

$$\begin{array}{cccccc} C_1 & V_1 & C_2 & V_2 & C_3 & V_3 \\ | & | & | & & | & | \\ \mathrm{b} & \mathrm{i} & \mathrm{t} & & \mathrm{ch} & \mathrm{i} \end{array}$$

2.7 Remaining Verbal Stems

- Periphrastic: *ootoba-t-il-ka* “we dream”
- Causative: *aaba-chi-l-ka* “we teach”

★ This dervies the positive agreement paradigms!

3 Derivations in Strict CV: Negative Agreement

- The negation paradigm always suffixed with the auxiliary *o*, which triggers vowel deletion of the previous vowel:
  - e.g. *choopa* → *cho<chi-kii>pa* → *cho<chi-kii>pa + o* → *cho<chi-kii>po* → *cho<chi-kii>po+ti*
- The derivations here do not include the negative auxiliary *o* and TAM morphology, for space reasons.

3.1 CVCCV

hokcho → *hochikikcho*

(11) a. Underlying form *hokcho*

C_1	V_1	C_2	V_2	C_3	V_3
h	o	k		ch	o

b. Insert negation and agreement material

C_1	V_1	<C_4	V_4	C_5	V_5>	C_2	V_2	C_3	V_3
h	o	ch		k		k		ch	o

c. All ungoverned slots filled at once

C_1	V_1	<C_4	V_4	C_5	V_5>	C_2	V_2	C_3	V_3
h	o	ch	i	k	i	k		ch	o

- Note that this does suggest that the “ungoverned” slots all at once, to ensure that V_4 is filled

3.2 CVVCV

(12) a. **1. Underlying form *choopa***

C_1	V_1	C_2	V_2	C_3	V_3
ch	o			p	a

b. **2. Infixation of NEG, right of V_1**

C_1	V_1	<C_4	V_4>	C_2	V_2	C_3	V_3
ch	o	k				p	a

c. **3. Infixation of 2SG, left of C_4**

C_1	V_1	<C_5	V_5>	<C_4	V_4>	C_2	V_2	C_3	V_3
ch	o	**ch**		**k**				p	a

d. **4. *i*-insertion:** V_5 and V_4 are ungoverned

C_1	V_1	<C_5	V_5>	<C_4	V_4>	C_2	V_2	C_3	V_3
ch	o	**ch**	i	**k**	i			p	a

e. **5. Remaining affixes & surface phonology:** *cho<chi-kii>p-o-ti* "you didn't buy'

3.3 CVCV

aɬi → *aɬchiko*

(13) a. *aɬi*

C_1	V_1	C_2	V_2
	a	ɬ	i

b. Agent agreement affixation

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4
	a	ɬ	i	ch		k	

c. *i*-insertion in ungoverned slot

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4
	a	ɬ	i	ch	i	k	

3.4 iCV

ila → *chiklo*

(14) a. Underlying form *ila*

C_1	V_1	C_2	V_2
		|	|
		l	a

b. NEG and AG insertion

C_1	V_1	<C_3	V_3	C_4	V_4>	C_2	V_2
		|		|		|	|
		ch		k		l	a

c. *i*-insertion into ungoverned slots

C_1	V_1	<C_3	V_3	C_4	V_4>	C_2	V_2
		|	|	|		|	|
		ch	i	k		l	a

3.5 -ka verbs

loska → *loschikko*

(15) a. Underlying *loska*

C_1	V_1	C_2	V_2	C_3	V_3	C_4	V_4
|	|	|			|	|	|
l	o	s			∅	k	a

b. <AG NEG> insertion

C_1	V_1	C_2	V_2	C_3	V_3	<C_5	V_5	C_6	V_6>	C_4	V_4
|	|	|			|	|		|		|	|
l	o	s			∅	ch		k		k	a

c. *i*-insertion

C_1	V_1	C_2	V_2	C_3	V_3	<C_5	V_5	C_6	V_6>	C_4	V_4
|	|	|			|	|	|	|		|	|
l	o	s			∅	ch	i	k		k	a

d. ∅ material is deleted

- Note that by comparing the positive and negative agreement derivations, the *i*-insertion for un-governed slots occurs twice:
  - It is evaluated *before* the ∅ material is deleted, shown in (15)-c
  - It is evaluated *after* the ∅ material is deleted, shown in (9)-c

3.6 Remaining Verbal Stems: *-li* and *-chi* Verbs

- The remaining verbs are *-li*, *-chi* verbs
- **Todo:** I need to still provide derivations of these, but these are similar because they seem to involve periphrastic conjugation. See table 5.
- **Todo:** The other issue is the relative ordering of AG and NEG in 1SG.
  - In 1SG for all verbal stems the order of morphemes is <a - k > *except in CVVCV and CVCCV, which as <k a(a)>*
  - *cho<k-aa>po-ti* "we didn't buy'
  - *los<ak>ko-ti* "we didn't buy'
- **Todo:** Note that in 1SG, there is a *t* between the verbal stem and agreement marker *aɬ<tak>ko*
  - This could suggest that CVCV verbs are actually periphrastic, and the /t/ only surfaces when the following segment is a vowel

4 Lexically Specified Conjugation Classes

- A small set of verbs belong to lexically specified conjugation classes
  - *nochi* 'to sleep'
  - *hiicha* 'to see'
  - *haalo* 'to hear'
- These are CVCV/CVVCV stems but take with the iCV conjugation, as illustrated in Table 1.

	SG	**PL**
1	nochi-li-ti	**il**-nochi-ti
2	**is**-nochi-ti	**has**-nochi-ti
3	nochi-ti	(ho)-nochi-ti

Table 1: *nochi* ‘to sleep’

(16) a. *nochi*

C_1	V_1	C_2	V_2	C_3	V_3
		n	o	ch	i

b. *il-nochi*

C_1	V_1	$<C_3$	$V_3>$	C_2	V_2	C_3	V_3
		l		n	o	ch	i

c. *i*-insertion for “monosyllabic” stem

C_1	V_1	$<C_3$	$V_3>$	C_2	V_2	C_3	V_3
	i	l		n	o	ch	i

d. melodic material linked (at some stage)

C_1	V_1	$<C_3$	$V_3>$	C_2	V_2	C_3	V_3
	i	l		n	o	ch	i

5 Remaining Issues

- **CCC Clusters**
- There are still some situations with CCC clusters: *bit<k>ko* “S/he didn’t dance”
- Compare this with the 3SG negation of *ka* verbs: *los<ik>ko* “s/he didn’t lie”
  - This suggests that there’s something about *li* verbs that stops the *i* insertion.

- **Suppletive Allomorphy Unexpected in 1PL Negation**
- Under the assumptions adopted here, suppletive allomorphy should already have been determined at this stage of the derivation, which would lead us to expect only the form in (17-b) instead.

(17) a. bit<kil>ko-ti
“We didn’t dance”

b. bit<hilik>ko-ti
“We didn’t dance”

- **Suppletive Roots**
- Many motion verbs in Alabama exhibit suppletive roots conditioned by number. For example, *aɬɬa* ‘to go’ shows distinct stems in singular, dual, and plural contexts, as illustrated in table 2, and *wiika / aswa / iisa* ‘to exist’, shown in table 3

	SG	DU	PL
1	aɬɬa-**li**-ti	a**n**ɬiiya-ti	amaa-**il**-ka-ti
2	aɬ-**chii**-ya-ti	aɬa-**hachii**-ya-ti	amaa-**s**-ka-ti
3	aɬɬa-ti	aɬɬa-**chi**-ti	amaa-ka-ti

Table 2: *aɬɬa* ‘to go’

	SG	DU	PL
1	wiika-**li**-ti	a-**li**-swa-ti	iis(a)-t-**il**-ka-ti
2	wii-**his**-ka-ti	a-**hachi**-swa-ti	iis(a)t-**as**-ka-ti
3	wiika-ti	aswa-ti	iisa-ti

Table 3: *wiika/aswa/iisa* ‘to exist’

Syll. Structure	Infix Placement	Example Word	1PL Output	2SG Output	2PL Output
i.CV	<iX>CV	*i.pa*	*il-pa*	*is-pa*	*has-pa*
-VVka	-VV<(h)iX>ka	*afaa.ka*	*afaa<(h)il>ka*	*afaa<(h)is>ka*	*afaa<has>ka*
-VCka	-VC<iX>ka	*los.ka*	*los<il>ka*	*los<is>ka*	*los<has>ka*
Periphrastic	-t-X-ka	*ootoba*	*ootoba-t-il-ka*	*ootoba-t-is-ka*	*ootoba-t-as-ka*
CVV.CV	CVV<X>CV	*hi.naa.ta*	*hinaa<l>ta*	*hinaa<s>ta*	*hinaa<(h)as>ta*
-chi	-chi<X>ka	*aaba.chi*	*abaa-chi<l>ka*	*aaba-chi<s>ka*	*abaa-ch<as>ka*
CV.CV	CVC<X>	*hom.pa.ni*	*hompan-hili*	*hompan-chi*	*hompan-hachi*
-li	-Xi	*bit.li*	*bit-hili*	*bit-chi*	*bit-hachi*
-li	-Xi	*soo.li*	*soo-hili*	*soo-chi*	*soo-hachi*
CVC.CV	CV<Xi>CCV	*hokcho*	*ho kcho*	*ho<chi>kcho*	*ho<hachi>kcho*

Table 4: Positive Agent Infix placement, organized by agent allomorph

Syll. Structure	1SG	1PL	2SG	2PL	3SG
i.la	*<**ak**>lo*	*<**kil**>lo*	*<**chik**>lo*	*<**hachik**>lo*	*<**ik**>lo*
afaa.ka	*afaa<**??**>ko*				
*los.**ka***	*los<**ak**>ko*	*los<**kil**>ko*	*los<**chik**>ko*	*los<**hachik**>ko*	*los<**ik**>ko*
choopa	*cho<**k-aa**>po*	*cho<**ki-lii**>po*	*cho<**chi-kii**>po*	*cho<**hachi-kii**>po*	*cho<**kii**>po*
*aaba.**chi***	*abaa-ch<**ak**>ko*	*abaa-chi<**kil**>ko*	*abaa-chi<**chik**>ko*	*abaa-chi<**hachik**>ko*	*abaa-chi<**k**>ko*
aɬi	*aɬ<**tak**>o*	*aɬ<**kil**>o*	*aɬ<**chik**>o*	*aɬ<**hachik**>o*	*aɬ<**k**>o*
*bit.**li***	*bit<**tak**>ko*	*bit<**kil**>ko*	*bit<**chik**>ko*	*bit<**hachik**>ko*	*bit<**k**>ko*
		*bit<**hilik**>ko*		*bit<**hask**>ko*	
*soo.**li***	*soo<**tak**>ko*	*soo<**kil**>ko*	*soo<**chik**>ko*	*soo<**hachik**>ko*	*soo<**k**>ko*
		*bit<**hilik**>ko*			
hok.cho	*ho<**ka**>kcho*	*ho<**kili**>kcho*	*ho<**chi-ki**>kcho*	*ho<**hachi-ki**>kcho*	*ho<**ki**>kcho*

Table 5: Negative Stem placement patterns across syllable structures

Syll. Structure	Infix Placement	Example Word	Output
i.CV	<iX>CV	*i.pa*	***<il>****pa*
-VVka	-VV<(h)iX>ka	*afaa.ka*	*??*
-VCka	-VC<iX>ka	*los.ka*	???
Periphrastic	-t-X-ka	*ootoba*	*??*
CVV.CV	CVV<X>CV	*choopa*	*choo****<l>****pa*
-chi	-chi<X>ka	*aaba.chi*	*abaachi-t****<il>****ka*
CV.CV	CVC<X>	*ta!a*	*ta!-****<ka>***
-li	-Xi	*batli*	*bat-****<ka>***
-li	-Xi	*albitiili*	*albitii-****<ka>***
CVC.CV	CV<Xi>CCV	*hosso*	*ho**** ****sso*

Table 6: Middle Voice morphology placement patterns across syllable structures